

Chern++: computational results on positivity for Morin Thom polynomials

Abstract

We report on a machine-checked study of the two positivity conjectures for Morin singularities. The infrastructure computes $\mathcal{Q}_d$ for $d \leq 6$ from the explicit orbit equations of Bérczi–Szenes §7.3, self-checked against every invariant fixed in advance. On top of it: Rimányi’s conjecture is verified for A_6 (first time), the two reductions of the A_5 handoff note are shown to survive at $d = 6$ while the proposed prefix-positivity route is shown to *fail* at $d = 6$, and a notion of *denominator certificate* is introduced whose successes are proofs and whose failures are theorems. The certificate machinery reproduces the $d = 4$ strong positivity theorem and Proposition 3 of the handoff note mechanically. All coefficients agree with Rimányi’s independently computed tables at relative dimension one. It does not close the A_6 tail, and we show this is a limitation of the technique rather than of the search: after generalising Lemma 1 to an absorption criterion, 13 090 candidate certificates fail. Notation follows `rimanyi_positivity.pdf`, `report.pdf` and `a5_weak_positivity_handoff.pdf`, cited as [N], [R], [H].

1 Setting and notation

For the Morin singularity A_d , the Bérczi–Szenes formula expresses Tp_{A_d} as an iterated residue whose only d -dependent input is $\mathcal{Q}_d = \mathrm{mdeg}_{T_d}(\mathcal{O}_d)$, the equivariant multidegree of the Borel orbit closure $\mathcal{O}_d = \overline{B_d \cdot \epsilon_{\mathrm{ref}}}$ inside $\widehat{N}_d \subset \mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2 \mathbb{C}^d)$. Normalising $z_d = 1$ and setting $x_j = z_j/z_{j+1}$, positivity is read off the chamber series

$$F_d(x) = \frac{\prod_{m < l} (1 - z_m/z_l) \mathcal{Q}_d}{\prod_{m+r \leq l} (1 - z_m/z_l - z_r/z_l)} = \frac{N_d(x)}{\prod_r (1 - f_r(x))} = \sum_{\beta \geq 0} A_\beta x^\beta,$$

where every f_r has nonnegative coefficients and zero constant term. Write $\tau(i, j, k, \dots) = (j - i, j, k, \dots)$ for the first adjacent transposition, and $C(M)$ for the ℓ -free Chern coefficient of [R, Thm. 5].

2 Infrastructure

2.1 Computing $\mathcal{Q}_d$

We take the route of [N, §7.3]: rather than parametrise the orbit and eliminate the group variables, we write down the quadratic relations of Proposition 7.3 directly, saturate by the defect-zero coordinates to remove the spurious components that appear from $d = 6$ on ([N, §7.4]), and read the multidegree off an initial ideal. Runtime for $d \leq 6$ is seconds.

The design point worth stating is that every quantity the computation can be checked against is known in advance, and all of them are checked: multihomogeneity of each relation under the torus weights; the codimension $\dim \widehat{N}_d - \binom{d}{2}$ forced by homogeneity of the residue formula; vanishing on random B_d -translates of ϵ_{ref} ; $\deg \mathcal{Q}_d$; exactness of the chamber division; and the normalisation fixing the c_1^d coefficient. The stage refuses to emit an artifact otherwise. Externally, all eleven classical A_6 coefficients are reproduced.

2.2 Scale of the objects

Table 1 is the practical reason sampling is no substitute for a proof: roughly half the coefficients of N_d are negative at every order, and the counts grow fast.

Table 1: The mined chamber algebras. $\deg \mathcal{Q}_d = \dim \hat{N}_d - \binom{d}{2}$ is the codimension of the orbit closure.

	vars	$\dim \hat{N}_d$	$\deg \mathcal{Q}_d$	$\mathcal{Q}_d$ terms	N_d terms	$\deg N_d$	N_d negative
A_4	3	7	1	3	54	13	27 (50%)
A_5	4	13	3	19	814	31	414 (51%)
A_6	5	22	7	418	26618	66	13309 (50%)

2.3 Exactness

No result below depends on floating point. One point deserves emphasis because it is a trap: Laurent coefficients are accumulated in `int64`, and they outgrow it. A wrapped accumulator reads as a large *negative* number, that is, as a spurious counterexample to the very conjecture under test. The evaluator therefore replays the fixed point in floating point as a magnitude probe and raises rather than returning wrapped values. Beyond that point the expansion is run in residue arithmetic instead; see §3.

3 Results for A_6

Proposition 1 (verified). *Rimányi’s conjecture holds for A_6 at every relative dimension $\ell \leq 7$, and for A_5 at $\ell \leq 11$.*

These are finite exact computations, not proofs of the infinite statement. For A_5 this matches the range of [R, Thm. 8]; the A_6 verification is new. Past $\ell = 5$ the coefficients outgrow `int64` — they reach 1.75×10^{22} at A_6 , $\ell = 7$ — so the expansion is run modulo several word-sized primes and reconstructed by CRT, with one prime held back and spent on verifying the reconstruction. The grouping into Chern monomials is computed from the grid geometry rather than from which cells are nonzero, so that the residues of a given coefficient correspond across primes; grouping by value would misalign them, since a nonzero coefficient can vanish modulo one prime and not another.

3.1 External agreement

Rimányi’s tables at `tpp.web.unc.edu` give the Thom polynomials at relative dimension one, computed by the restriction-equation method. All 15, 30 and 58 coefficients for A_4 , A_5 , A_6 agree with ours. Since that method is independent of the residue formula implemented here, this constrains $\mathcal{Q}_d$ externally rather than self-consistently, and is a considerably sharper check than the eleven classical values at $\ell = 0$.

The negative Laurent coefficients of F_6 behave much as at $d = 5$; see Table 2. The first counterexample to strong Laurent positivity occurs at $d = 6$ at the same β as at $d = 5$, extended by a zero. The last column is the one that matters in §3.3.

3.2 The reductions of $[H]$ at $d = 6$

Observation 2. *Both reductions of $[H]$ survive at $d = 6$ throughout the tested range ($\deg \leq 11$): the unpaired tail $i > j \Rightarrow A_{i,j,\dots} \geq 0$, and the paired inequality $A_{i,j,\dots} + A_{j-i,j,\dots} \geq 0$ for $0 \leq i \leq j$.*

This is evidence that the τ -pairing frame is structural rather than an A_5 accident.

Table 2: Negative Laurent coefficients of the chamber series, to total degree 11.

	terms	negative	first negative	most negative	with $i = 0$
A_4	237	0	—	—	0
A_5	490	6	(1, 1, 2, 1)	-1	0
A_6	796	10	(1, 1, 2, 1, 0)	-2	2

3.3 Prefix positivity fails at $d = 6$

[H, §8] observes that every tested coefficient of $B = F_5/(1 - a)$ is nonnegative, and [H, §11.4] proposes proving that as a stepping stone, since $B_{i,j,k,l} = \sum_{r \leq i} A_{r,j,k,l}$ turns the paired inequality into a statement about adjacent differences of a prefix array.

Proposition 3. $F_6/(1 - a)$ is not coefficientwise nonnegative. Indeed $A_{(0,2,3,2,2)} = -1$.

Proof. The prefix sum runs over $r \leq i$, so $B_{0,j,k,l} = A_{0,j,k,l}$ and any negative coefficient with $i = 0$ survives untouched. F_5 has no negative coefficient with $i = 0$ in the tested range; F_6 has two, the first being $A_{(0,2,3,2,2)} = -1$. Confirmed along two independent code paths, truncated series products in exact integers and the XLA fixed-point grid. $\square$

The route of [H, §11.4] is therefore specific to $d = 5$. Note that the obstruction is at the very first level of the prefix sum, so no refinement of the argument recovers it.

4 Denominator certificates

Definition 4. Let $N/\prod_{r=1}^R(1 - f_r)$ with each f_r nonnegative and of zero constant term. An *order- k certificate* is an identity

$$N = \sum_{|S| \leq k} P_S \prod_{r \in S} (1 - f_r), \quad \text{every } P_S \text{ coefficientwise nonnegative,}$$

over subsets $S \subseteq \{1, \dots, R\}$.

Dividing through, $F = \sum_S P_S / \prod_{r \notin S} (1 - f_r)$ is a sum of products of nonnegative series, so $F \geq 0$. Order 0 is the statement that N itself is nonnegative. [H, §10.3] reports a first-order search of this shape coming back infeasible.

4.1 Why a returned certificate is a proof

$P_\emptyset$ is not a free unknown: since $\prod_{r \in \emptyset} (1 - f_r) = 1$, it is the remainder $N - \sum_{S \neq \emptyset} P_S \prod_{r \in S} (1 - f_r)$. We therefore solve a feasibility LP in the nonempty parts only, snap the solver’s floating-point answer to exact rationals, and recompute $P_\emptyset$ in exact arithmetic. The identity then holds by construction, and the single remaining question, whether that exact remainder is coefficientwise nonnegative, is decided exactly. The LP is demoted to a search heuristic; correctness never depends on it. The output is a finite algebraic identity checkable in any computer algebra system.

4.2 Why a failure is a theorem

Lemma 5 (projection). *A constraint on a monomial of total degree T involves only the coefficients of the P_S of degree $\leq T$, because every $\prod_{r \in S} (1 - f_r)$ has constant term 1 and no negative-degree terms. Truncating both the constraints and the unknowns at degree T is therefore an exact projection of the feasible set, not a further restriction.*

Consequently infeasibility of the truncated program proves that no certificate of that order exists at *any* degree. Failed searches become impossibility results.

4.3 A_4

Proposition 6. F_4 admits an order-4 certificate with $\deg P_S \leq 8$, exhibited explicitly and verified exactly; and no certificate of order ≤ 3 exists at any degree. The minimal order is 4.

This is an independent computational proof of [N, Thm. 1] by a method using no special feature of $d = 4$. The lower bound already follows by hand at depth 1: writing m_S for the constant term of P_S , matching the constant term of $N = 1 - a - b - c + \dots$ forces $\sum_S m_S \leq 1$, while matching the coefficients of a, b, c forces

$$\sum_{S \ni f_0} m_S \geq \frac{1}{2}, \quad \sum_{S \ni f_2} m_S \geq 1, \quad \sum_{S \ni f_5} m_S \geq 1,$$

where $f_0 = 2a$, $f_2 = b + ab$, $f_5 = c + abc$ are the only factors containing the respective bare monomials. With a budget of 1 these can only be met if a single S contains all three. The machine-found certificate saturates all four bounds exactly, its only two constant terms being $\frac{1}{2}(1 - f_0)(1 - f_2)(1 - f_4)(1 - f_5)$ and $\frac{1}{2}(1 - f_2)(1 - f_4)(1 - f_5)(1 - f_6)$.

4.4 A_5 : how far the obstruction reaches

Proposition 7. No denominator certificate of order ≤ 7 exists at any degree for F_5 , nor for $F_5/(1 - a)$.

For F_5 this is expected, since $F_5 \not\geq 0$, and serves as a control that the engine never manufactures a certificate for a false statement. For the prefix series, which *is* coefficientwise nonnegative as far as we compute, it is informative: it extends [H, §10.3] from order 1 to order 7. The bound sharpens with probe depth: depth 1 gives ≥ 4 , depth 2 gives ≥ 7 , depth 3 rules out everything up to 7.

5 The unpaired tail

5.1 One series

Set $H(a) = (1 - a)/(1 - 2a)$ and $J_d = F_d/H$. After H is removed, every occurrence of a in a numerator or denominator monomial is accompanied by b : the chamber monomials containing a are $z_1/z_l = ab \cdots$, and only $z_1/z_2 = a$ is bare, which is exactly the factor $1 - 2a$ that H removes. Hence $\deg_a [b^j c^k \cdots] J_d \leq j$, and convolving with $H(a) = 1 + \sum_{n \geq 1} 2^{n-1} a^n$ gives, for $i > j$,

$$A_{i,j,k,\dots} = 2^{i-1} [b^j c^k \cdots] J_d(1/2, b, c, \dots).$$

The entire unpaired tail, infinitely many coefficients, is therefore one positivity question about a single series in $d - 2$ variables. This is [H, Lemma 2 and Prop. 3] for $d = 5$, stated so that it applies verbatim for any d .

Our $J_5(1/2, b, c, e)$ agrees with [H, eq. (9)] coefficient for coefficient to degree 12, which is what validates the construction. $J_6(1/2, b, c, d, e)$ is coefficientwise nonnegative as far as we compute ($\deg \leq 10$).

5.2 Multiplicative certificates

The two successful proofs in the literature, [N, Thm. 1] and [H, Prop. 3], are *multiplicative*: products of ratios $(1 - u)/(1 - v)$, each nonnegative by Lemma 1 because $v - u \geq 0$. That shape is not a finite sum $\sum_S P_S \prod_{r \in S} (1 - f_r)$ with $P_S \geq 0$, so the additive certificates of §4 structurally cannot find them.

The diagnostic is decisive: the additive search is obstructed on $J_5(1/2, \cdot)$, a series whose nonnegativity *is* Proposition 3. When a method fails on a known theorem, the method is wrong.

Presenting the numerator in factored form, the Vandermonde part being already a factorisation, one can search for a matching of numerator factors to denominators satisfying the Lemma 1 condition, cancel whatever divides exactly, and ask whether what remains is nonnegative.

Proposition 8. *For $d = 5$ this closes: all nine numerator factors pair off, three denominators cancel exactly, and the leftover numerator is exactly 1. Proposition 3 is reconstructed mechanically.*

For $d = 6$ it does not. A maximum matching leaves

$$\frac{1 - cde - \frac{3}{2}bcde}{1 - 2cde},$$

whose expansion has the negative family $b(cde)^k$, and no available denominator v dominates $u = cde + \frac{3}{2}bcde$. Maximum matching is in fact greedy in the wrong way: it consumes denominators the remainder needs. Returning two ratios to the remainder leaves twelve Lemma-1 ratios times a remainder (207-term numerator over nine denominators) that is nonnegative as far as we check.

5.3 Generalising Lemma 1, and a negative result

Lemma 1 requires the numerator in the shape $1 - u$. The following drops that and lets several denominators cooperate.

Lemma 9 (absorption). *Write $N = N_+ - N_-$ with both parts coefficientwise nonnegative, and let S be a set of denominators with $1 - W = \prod_{v \in S} (1 - v)$. If $N_- \leq N_+ \cdot W$ coefficientwise, then $N / \prod_{\text{all}} (1 - v)$ is coefficientwise nonnegative.*

Proof. $N - N_+ \prod_S (1 - v) = N_+ W - N_- \geq 0$, so $N = N_+ \prod_S (1 - v) + P$ with $P \geq 0$ and

$$\frac{N}{\prod_{\text{all}} (1 - v)} = \frac{N_+}{\prod_{v \notin S} (1 - v)} + \frac{P}{\prod_{\text{all}} (1 - v)},$$

both nonnegative. □

Lemma 1 is the case $N = 1 - u$, $S = \{v\}$, where the condition reads $u \leq v$; the criterion is strictly stronger, since it applies to numerators not of that shape.

Observation 10. *Searching jointly over back-off subsets and absorbing subsets, 13 090 candidate criteria, closes nothing for A_6 .*

Since $J_6(1/2, \cdot)$ is nonnegative as far as we compute, a proof exists. The content of the observation is that it is not of this form: the technique that proves $d = 5$ does not extend, and the obstruction is structural rather than a matter of searching harder. Concretely, the negativity of the residual factor is compensated by the *other* ratios, so no local pairing argument can succeed.

6 Assessment

What the computations suggest about proof strategy:

1. The τ -pairing reduction is the right frame, both its inequalities surviving at $d = 6$, but the prefix-positivity stepping stone is not, and is unavailable from $d = 6$ on.
2. Additive, Positivstellensatz-shaped certificates are obstructed on *every* target tried, including provable ones. On present evidence they are not the tool for this problem, and we would not recommend further effort extending [H, §10.3] in that direction.

3. Multiplicative certificates are the tool that works, and they are now mechanised: they reproduce both published proofs from scratch. Their reach stops before A_6 .
4. The natural remaining targets are the paired series $\mathcal{P}$ of [H, eq. (12)], for which the construction is now cheap since $S = J_5(1/2, \cdot)$ is exactly the object above, the reflection is an index reflection, and [H, §7] supplies the denominator; and closed-form domination along the negative families of [R, §7.2], where each family settled is infinitely many cases.

Reproducibility

Everything above is reproducible from a clean checkout of Chern++. Statements are separated throughout into finite exact verifications and proofs; the certificate machinery and the closed-form family arguments are the only components producing the latter. The annotated walkthrough is `src/examples.ipynb`, executed with its outputs.